Course: CSE303 (Statistics for Data Science)

Lab Manual 01: Set up your system

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Lab Report : 01

Submitted To:

Puja Chakraborty

Lecturer, East West University

Submitted By:

Afsin Sultana

ID : 2022-1-60-113

1. Describe what is python script, python module and library?

Answer:

Python Script is a file where set of instruction contain with some specific task or line of code .

A file containing a set of functions you want to include in your application . To create a module just save the code you want in a file with the file extension .py .

Python is a high-level programming language where we have to use python standard library .It plays very important role.There are some python standard library for example , tensorflow ,matplotlib ,pandas ,

Numpy,Scipy, Scrapy , pygame , pytorch ,django and so on. It is essential tool that provide reuseable code to perform various task efficiently.

2. Point out the difference and use of .py and .ipynb files.

Answer:

.py file:

* With a .py file, one will be able to test individual components of the pipeline more easily.

* It allows for a more traditional development workflow, where someone can write and run tests using frameworks like pytest.

*.py files are easier to debug with IDEs like VS Code, making it easier to find errors.

*Execution of .py files starts fresh, unlike .ipynb files where some variables can get carried over from the last execution or deleted cells.

.ipynb file:

*With a .ipynb file, one will be able to check the data while creating the pipeline, resulting in a better developer experience.

*It provides an interactive environment where I can explore and visualize data.

Some tools like Databricks have a notebook interface for creating pipelines.

In jupyter notebook we can use .ipynb files.

.py is a regular python file. It's plain text and contains just your code.

On the other hand , .ipynb is a python notebook and it contains the notebook code, the execution results and other internal settings in a specific format. You can just run .ipynb on the jupyter environment.

3. What is VsCode and Anaconda?
   Answer:
   VS code is the ide where one can code whereas anaconda is a tool to manage python version ,virtual environment and package or modules

4. Differentiate between conda and Anaconda.
   Answer:
   Conda is the open source package management system that helps developer install , manage , update packages dependencies on various language whereas anaconda is other scientific package that aims to process python environment for data science and scintific computing. Anaconda is wide comprehensive in nature. Conda provides command line interface for user but anaconda includes a graphical user interface (GUI) called Anaconda Navigator, which

offers a more user-friendly approach to managing environments and packages. While Conda provides a lightweight package management system, Anaconda offers a complete distribution with additional pre-installed packages.

5.Do you need to install python separately if you install Anaconda?

Answer:

No , I do not need to install Python separately when using Anaconda, as Anaconda comes with its own version of Python.

6. What is a python environment?

Answer:

A Python environment is a self-contained directory that contains a specific version of Python and a collection of packages and dependencies. This setup allows developers to manage and isolate project dependencies, ensuring that each project has its own set of libraries and does not interfere with other projects or the system Python installation.

7. How to integrate conda with VS Code?

Answer:

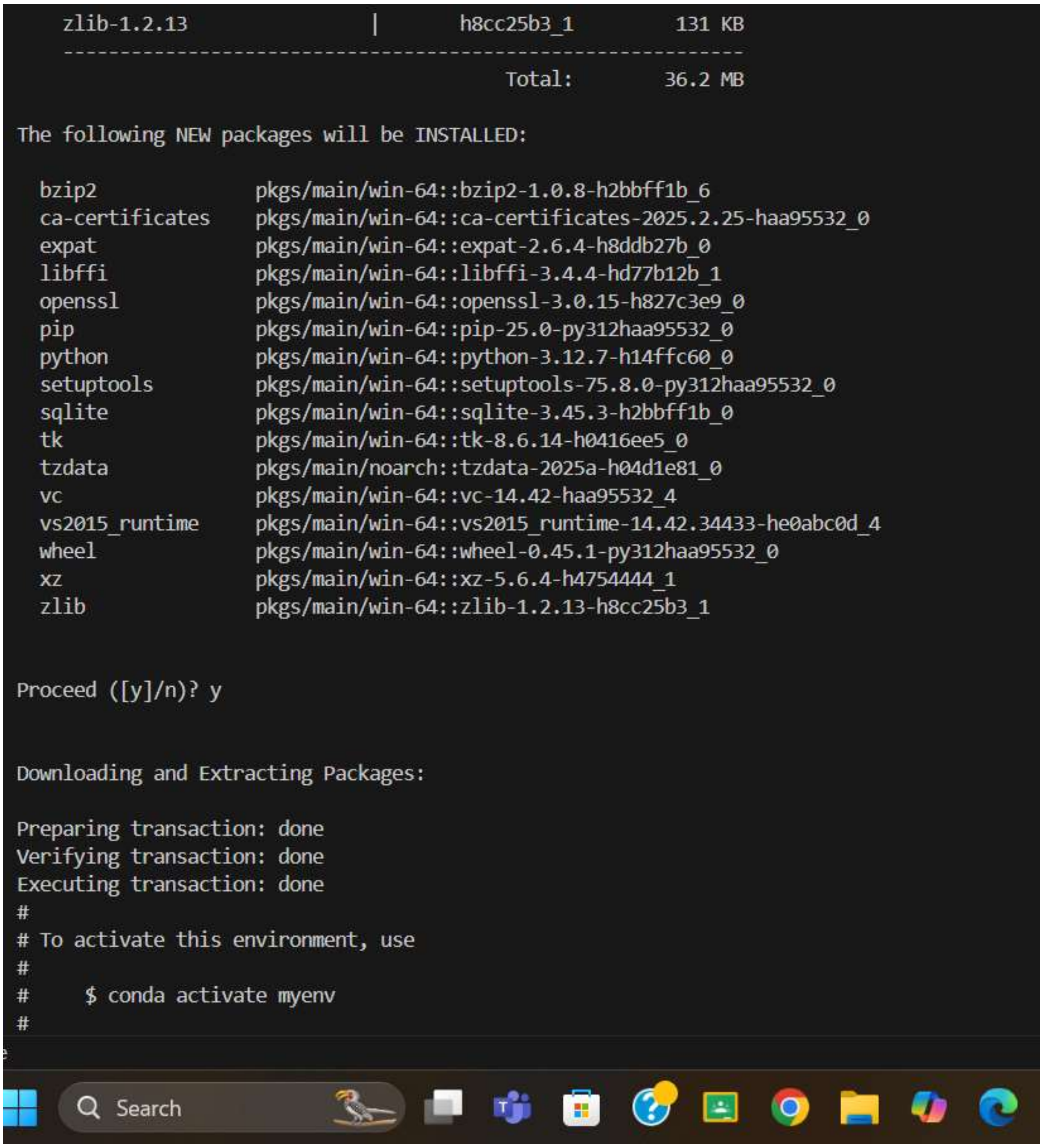
We have to use test environment to use conda.

8 .no question answer :

a) We have to write conda env list

B) conda create - - name myenv python = 3.12.7

```
PS D:\CSE303_stat_for_DS> conda create --name myenv python=3.12.7
Channels:
 - defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: C:\Users\ARPITA\Downloads\anaconda\envs\myenv

  added / updated specs:
    - python=3.12.7


The following packages will be downloaded:

    package                    |            build
    ---------------------------|-----------------
    bzip2-1.0.8                |       h2bbff1b_6          90 KB
    expat-2.6.4                |       h8ddb27b_0         257 KB
    libffi-3.4.4               |       hd77b12b_1         122 KB
    openssl-3.0.15             |       h827c3e9_0         7.8 MB
    pip-25.0                   |  py312haa95532_0         3.0 MB
    python-3.12.7              |       h14ffc60_0        16.4 MB
    setuptools-75.8.0          |  py312haa95532_0         2.2 MB
    sqlite-3.45.3              |       h2bbff1b_0         973 KB
    tk-8.6.14                  |       h0416ee5_0         3.5 MB
    tzdata-2025a               |       h04d1e81_0         117 KB
    vc-14.42                   |       haa95532_4          11 KB
    vs2015_runtime-14.42.34433 |       he0abc0d_4         1.2 MB
    wheel-0.45.1               |  py312haa95532_0         177 KB
```



```
    zlib-1.2.13                |       h8cc25b3_1         131 KB
    ------------------------------------------------------------
                                           Total:        36.2 MB

 The following NEW packages will be INSTALLED:

   bzip2              pkgs/main/win-64::bzip2-1.0.8-h2bbff1b_6
   ca-certificates    pkgs/main/win-64::ca-certificates-2025.2.25-haa95532_0
   expat              pkgs/main/win-64::expat-2.6.4-h8ddb27b_0
   libffi             pkgs/main/win-64::libffi-3.4.4-hd77b12b_1
   openssl            pkgs/main/win-64::openssl-3.0.15-h827c3e9_0
   pip                pkgs/main/win-64::pip-25.0-py312haa95532_0
   python             pkgs/main/win-64::python-3.12.7-h14ffc60_0
   setuptools         pkgs/main/win-64::setuptools-75.8.0-py312haa95532_0
   sqlite             pkgs/main/win-64::sqlite-3.45.3-h2bbff1b_0
   tk                 pkgs/main/win-64::tk-8.6.14-h0416ee5_0
   tzdata             pkgs/main/noarch::tzdata-2025a-h04d1e81_0
   vc                 pkgs/main/win-64::vc-14.42-haa95532_4
   vs2015_runtime     pkgs/main/win-64::vs2015_runtime-14.42.34433-he0abc0d_4
   wheel              pkgs/main/win-64::wheel-0.45.1-py312haa95532_0
   xz                 pkgs/main/win-64::xz-5.6.4-h4754444_1
   zlib               pkgs/main/win-64::zlib-1.2.13-h8cc25b3_1


 Proceed ([y]/n)? y


 Downloading and Extracting Packages:

 Preparing transaction: done
 Verifying transaction: done
 Executing transaction: done
 #
 # To activate this environment, use
 #
 #     $ conda activate myenv
 #
```


c)

Conda activate myenv

```
# conda environments:
#
base                  *  C:\Users\ARPITA\Downloads\anaconda
myenv                    C:\Users\ARPITA\Downloads\anaconda\envs\myenv

PS D:\CSE303_stat_for_DS> conda activate myenv
PS D:\CSE303_stat_for_DS> 
```

9. What is extension (in VSCode)?

Answer:

VSCode Extensions are plug-ins developed by the community to enhance the functionality of Visual Studio Code (VSCode).

10. Create a project (folder/repository) named "CSE303" that should contain the following folders:

Answer: